

3DM 2024 Proof Workshop 2: Some additional questions

Let G be a graph. A *colouring* of G is an assignment of colours to the vertices of G so that adjacent vertices have different colours. A graph is *k -colourable* if it has a colouring with k or fewer colours. The *chromatic number* of G is

$$\chi(G) = \min\{k : G \text{ is } k\text{-colourable}\}.$$

So $\chi(G) \leq 2$ if and only if G is bipartite.

1. Determine $\chi(C_n)$.
2. Determine $\chi(K_n)$.
3. Prove that if $G \subseteq H$, then $\chi(G) \leq \chi(H)$.
4. Find the chromatic number of the Petersen Graph.
5. Prove that the graph G below has $\chi(G) = 4$.

